

LIMB BUDS

The limbs develop from the lateral body wall; their outgrowth is initiated at defined positions along the embryonic axis where cells continue to proliferate, giving rise to local thickenings that soon develop into limb buds. These buds are encased in an ectodermal epithelium rimmed by a longitudinal ridge of columnar epithelial cells: the apical ectodermal ridge. **Figure 15.1** shows the main stages in the development of a human upper limb; the stages in development of the lower limb are basically the same. The early limb bud elongates and, gradually, the different limb regions become apparent. A broad plate forms at the tip and, within it, digital rays develop that mark the position of the forming digits. The digits later separate and become tipped with nails. For further details of upper and lower limb development, see the appropriate regional chapters.

The early limb bud contains a mixed population of mesenchymal cells: somatopleuric mesenchyme cells that give rise to the connective tissues, including cartilage, bone, tendon, loose connective tissue and

dermis; and paraxial mesenchyme cells that have migrated from the somites, and which give rise to the myogenic cells of the muscles and to some of the endothelial cells that produce an extensive vascular network in the early limb bud. The ectodermal epithelium covering the limb bud gives rise to the epidermis of the skin. Motor and sensory nerves and their associated Schwann cells, together with melanocytes destined for the skin, migrate into the developing limb somewhat later; the Schwann cells and melanocytes are derived from the neural crest.

AXES OF LIMBS

For descriptive, experimental and conceptual purposes, various 'axes', borders, surfaces and lines in relation to the developing limb bud are defined and named (**Fig. 15.2**). An imaginary line from the centre of the elliptical base of the bud, through the centre of its mesenchymal core to the centre of the apical ectodermal ridge, defines the proximo-distal axis of the limb bud (previously known in descriptive embryology simply as the axis). Named in relation to the proximodistal axis, the

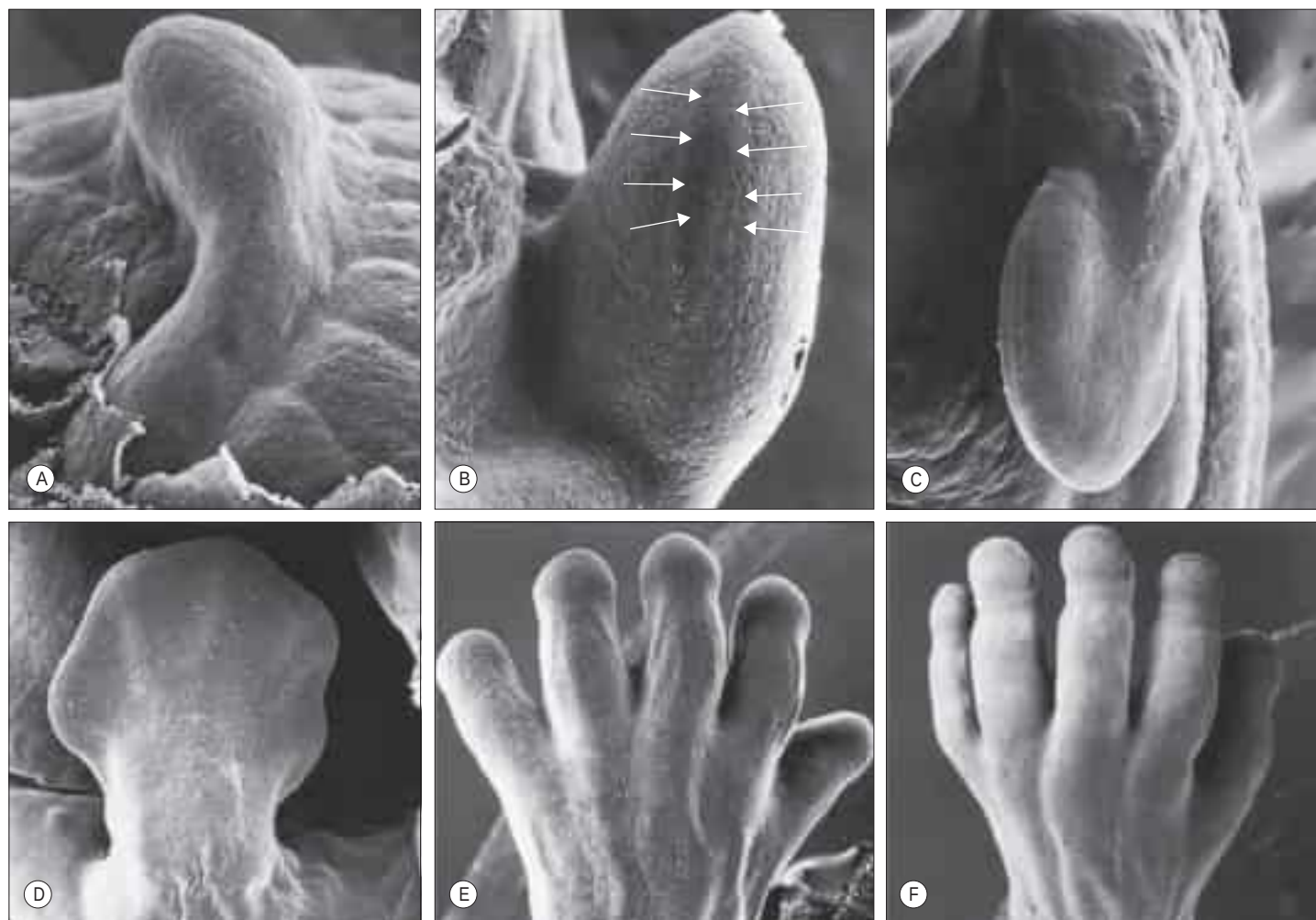


Fig. 15.1 Scanning electron micrographs to show the development of the upper limb. **A**, An early limb bud viewed from the postaxial border. **B**, A limb bud viewed from the postaxial border; the apical ectodermal ridge (arrows) can be seen. **C**, A limb bud, dorsolateral view; the shoulder and elbow region can be discerned, and a hand plate has formed. The apical ectodermal ridge is still obvious at the margin of the hand plate. **D**, Digital rays are present in the hand plate and the margin of the plate is becoming notched. **E**, The fingers are nearly separated and proliferation is commencing at the distal end of each digit to form the nail bed. **F**, The fingers each have tactile pads distally, and nail development continues. (Photographs courtesy of P Collins; printed by S Cox, Electron Microscopy Unit, Southampton General Hospital.)

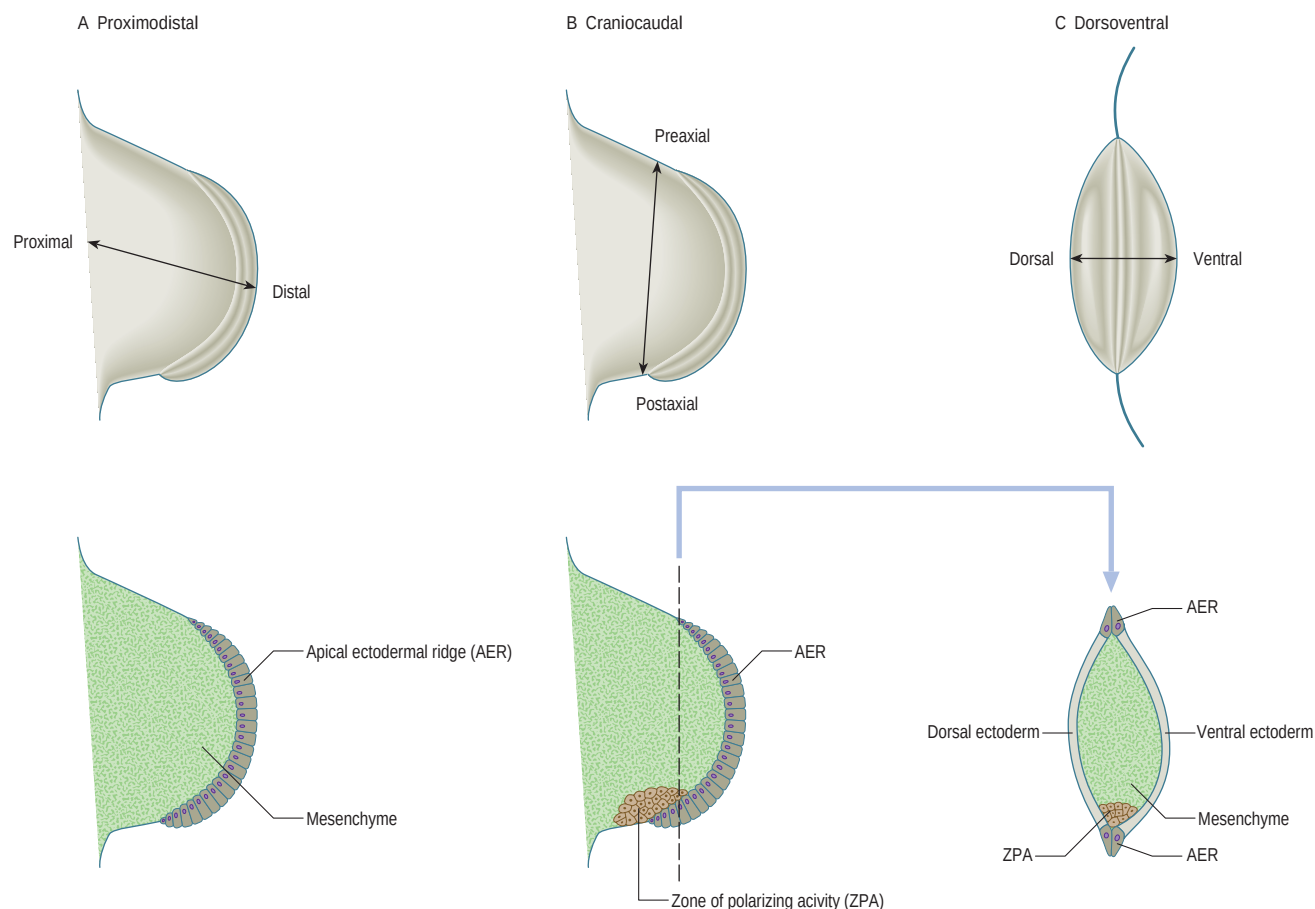


Fig. 15.2 Axes of the developing limb and specification of structures along the different axes. **A**, Proximodistal; limb bud viewed from the dorsal side. **B**, Craniocaudal; limb bud viewed from the dorsal side. **C**, Dorsoventral; limb bud viewed from the distal end. The development of limb structures and the ectodermal specializations are controlled by the limb mesenchyme. Outgrowth of limb bud is controlled by the apical ectodermal ridge and accompanied by the laying down of structures along the proximodistal axis; craniocaudal pattern is controlled by the zone of polarizing activity (polarizing region); dorsoventral pattern is controlled by signals from dorsal and ventral ectoderm, shown in cross-section; the section taken in the plane is shown by the dotted line. The limb bud in the diagram is equivalent to that of a human embryo at approximately stage 14.

cranially placed limb border is the preaxial border and the caudally placed limb border is the postaxial border. (In vertebrate embryos used as experimental models, e.g. chick embryos, the pre- and postaxial borders are termed anterior and posterior borders, respectively.) Any line that passes through the limb bud from preaxial to postaxial border, orthogonal to the proximodistal axis, constitutes a craniocaudal axis. The dorsal and ventral ectodermal surfaces clothe their respective aspects from preaxial to postaxial borders, and any line that passes from dorsal to ventral aspect, orthogonal to both proximodistal and craniocaudal axes, constitutes a dorsoventral axis. It should be noted here that the terms dorsal and ventral axial lines are to be used exclusively in relation to developing and definitive patterns of cutaneous innervation of the limbs and their associated levels of the trunk.

The three developmental axes (proximodistal, craniocaudal and dorsoventral) can be identified in the human developing limb bud by stage 13. Experiments on chick embryos have shown that a different set of cell-cell interactions control the development of structures in relation to each of the three principal axes of the limb bud: proximodistal (e.g. humerus to fingers), craniocaudal (e.g. thumb to little finger) and dorsoventral (e.g. extensors to flexors) (see Fig. 15.2). These experiments are described in detail in Hinchcliffe and Johnson (1980).

Outgrowth of the limb bud along its proximodistal axis is controlled by interactions between the apical ectodermal ridge and underlying somatopleuric mesenchyme, and is accompanied by the sequential formation of limb structures along the proximodistal axis. Growth and development along its craniocaudal axis involves an interaction between a small population of mesenchyme cells on the postaxial border of the limb bud, termed the zone of polarizing activity, or polarizing region, and adjacent mesenchyme. The development of the dorsoventral axis of the limb involves an interaction between the surface ectoderm on the sides of the limb bud and the underlying mesenchyme. These sets of cell-cell interactions are coordinated so that the developing limb develops properly with respect to all three axes.

Early differential growth of parts of the limb bud results in two main changes to the originally symmetrical axes of the limb. The dorsal aspect of the limb grows faster than the ventral, which causes the limb bud to

curve around the body wall; the ventral surface of the limb (closest to the body wall) remains relatively flat, but the dorsal surface bulges into the amniotic cavity. The apical ectodermal ridge, originally facing laterally, becomes increasingly directed caudally (upper limb) and ventrally (lower limb) as the limbs extend around the body.

Apical ectodermal ridge

The apical ectodermal ridge is required for limb bud outgrowth. When it is removed from a chick wing bud, outgrowth ceases and a truncated wing results, whereas when a second apical ectodermal ridge is grafted, a new outgrowth is induced that develops digits (Saunders et al 1976) (Fig. 15.3). The molecular basis of the cell-cell interactions between apical ridge and underlying mesenchyme is conserved between different vertebrates; the apical ectodermal ridge of a mouse limb bud can promote outgrowth of a chick wing bud.

There is evidence that the limb mesenchyme beneath the apical ectodermal ridge provides a factor that is essential for maintaining activity of the ridge (the so-called apical ridge maintenance factor). Thus, the craniocaudal extent of production of apical ectodermal ridge maintenance factor determines the length of the apical ectodermal ridge. When chick leg bud cells are grafted to the tip of a wing bud beneath the apical ectodermal ridge, the ridge is maintained and outgrowth continues. It should be noted that the leg cells, even though placed in a wing bud, still form distal leg structures, i.e. toes (Saunders and Gasseling 1959) (see Fig. 15.3). In addition, leg mesenchyme will pass information to local wing ectoderm, eliciting appropriate epidermal development: in this case, formation of scales rather than feathers.

For many years, it was thought that a timing mechanism linked to limb bud outgrowth specified structures along the proximodistal axis. The timing mechanism was postulated to operate at the tip of a developing limb bud beneath the apical ectodermal ridge in the zone of proliferating undifferentiated mesenchyme cells, which were therefore called the progress zone. It was assumed that, as cells left the progress

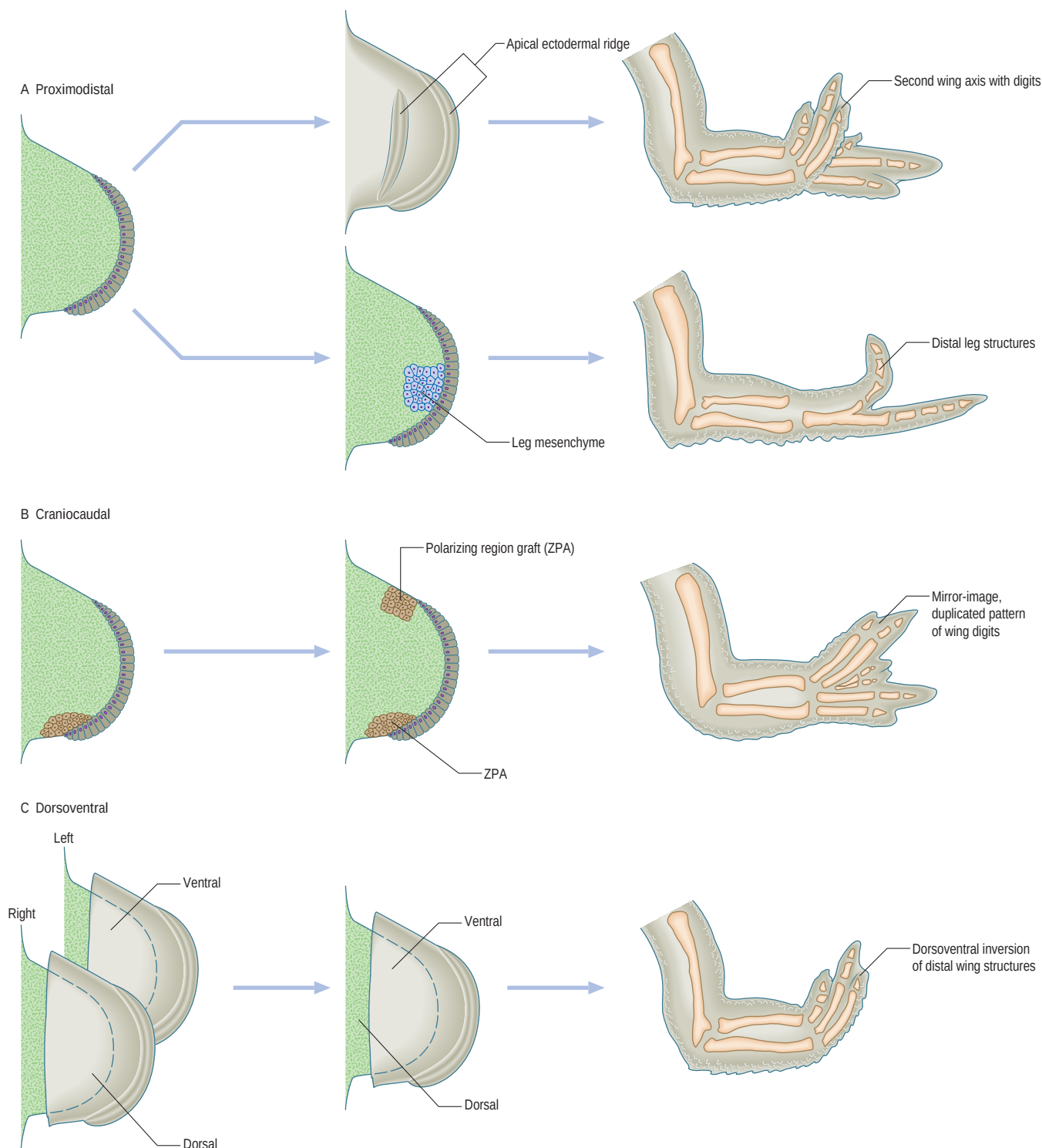


Fig. 15.3 A–C, Experimental manipulations that identified the fundamental cell–cell interactions involved in patterning the three main axes of the limb in the chick. Abbreviation: ZPA, zone of polarizing activity.

zone during limb bud outgrowth, their proximodistal fate became fixed. Thus, cells spending a short time at the tip of the limb bud would be specified to form proximal structures, while cells spending a longer time would be specified to form distal structures (Summerbell et al 1973). However, it now seems that the proximodistal pattern is specified in the very early limb bud (Dudley et al 2002) and that the prespecified structures are progressively elaborated as the bud grows out.

ZONE OF POLARIZING ACTIVITY

The zone of polarizing activity (or polarizing region) at the postaxial border was discovered by a grafting experiment on chick wing buds (Saunders and Gasseling 1968). When a polarizing region is grafted to the preaxial border of an early chick wing bud, so that there are polarizing region cells at both borders, duplication of distal limb structures occurs, with two ulnae (instead of an anterior radius and posterior

ulna) and a mirror-image pattern of six digits (the normal chick wing has three digits (see Fig. 15.3)). The digit closest to the polarizing region is always the most postaxial digit (in the chick wing, traditionally designated as digit 4), while more preaxial digits develop further away. Thus, the signal from the polarizing region appears to control both digit number and digit pattern. Digit number is controlled by regulating the width of the limb bud: both directly, via promoting cell proliferation, and indirectly, by controlling production of the apical ectodermal ridge maintenance factor. Digit pattern is specified by concentration of a diffusible morphogen produced by the polarizing region. The width of the limb bud is also limited by zones of programmed cell death on the preaxial and postaxial borders. (It should be noted that, in the literature relating to work on animal embryos, pre- and postaxial borders are termed anterior and posterior borders, respectively, and therefore the regions in which programmed cell death occurs are termed the anterior and posterior necrotic zones.) When the length of the apical ectodermal ridge is reduced, fewer digits form (oligodactyly), while, when the apical

ectodermal ridge becomes longer, more digits form (polydactyly). Other regions of apoptotic cell death occur between the digits and result in digital separation, but these appear later than the anterior and posterior necrotic zones; the resultant debris in all of the necrotic zones is removed by macrophages.

Grafts of the zone of polarizing activity only generate mirror-image digit patterns when placed beneath or adjacent to the apical ectodermal ridge, suggesting that the ridge produces a factor that maintains polarizing region activity. A positive feedback loop, where the polarizing region and the apical ectodermal ridge mutually maintain each other's activity, would ensure that development of craniocaudal and proximodistal axes is coordinated. Like that of the apical ectodermal ridge, the function of the polarizing region is conserved across species. Thus, grafts of the postaxial border of limb buds from mammals, including human limb buds (Fallon and Crosby 1977), generate additional chick digits, showing that the polarizing region produces the same molecular signal, although its interpretation depends on the responding cells.

ECTODERMAL INTERACTION

Signals from the ectoderm control the dorsoventral axis. It is possible to remove the surface ectodermal epithelium (like peeling off a glove) from the mesenchymal core of an early chick limb bud. When the ectodermal epithelium from a left limb bud is recombined with the mesenchymal core from a right limb bud, keeping the craniocaudal axis in register, the distal part of the limb that develops conforms to the polarity of the ectoderm rather than the polarity of the mesenchyme (MacCabe et al 1974) (see Fig. 15.3). This can be seen in the pattern of muscles, orientation of the joints and eventual characteristic differentiation of the epidermis.

MOLECULAR BASIS OF CELL-CELL INTERACTIONS

The apical ectodermal ridge produces fibroblast growth factors, extracellular cell-cell signalling molecules that act on the underlying mesenchyme cells and are essential for limb bud outgrowth (Niswander et al 1993). Fibroblast growth factors also maintain the activity of the polarizing region. The latter expresses *Sonic hedgehog* (*Shh*), a gene encoding another class of secreted protein that acts as an extracellular signalling molecule (Riddle et al 1993). *Sonic hedgehog* protein (*Shh*) acts on adjacent mesenchyme cells, controlling their proliferation and also maintaining the expression of the *Gremlin* gene, encoding an extracellular antagonist of secreted bone morphogenetic proteins. *Gremlin* functions as the apical ectodermal ridge maintenance factor (Zuniga et al 1999). *Sonic hedgehog* plays a pivotal role in the specification of the craniocaudal digit pattern: it spreads across the limb bud so that cells at different positions across the craniocaudal axis are exposed to different *Shh* concentrations (Tickle and Barker 2013). Responding cells transduce the signal through the *Gli* family of transcription factors. (The *Gli* proteins are bifunctional effectors of *Shh* signalling and can lead to either activation or repression of expression of target genes.) Dorsal limb ectoderm expresses *Wnt7a*, encoding another extracellular signalling molecule that acts as a dorsalizing signal (Parr and McMahon 1995).

Among the genes expressed in response to cell-cell signalling molecules in the limb are 5' members of the *Hox-a* and *Hox-d* complexes. *Hox* genes code for transcription factors that act in a cell-autonomous fashion to control expression of many target genes. *Hox-d* gene expression in the distal limb bud shows a nested arrangement similar to that of Russian babushka dolls: the postaxial border of the limb bud tip expresses five *Hox-d* genes (*d-13*, *d-12*, *d-11*, *d-10* and *d-9*), while in the adjacent, more preaxial, portion, only four genes are expressed (*d-12*, *d-11*, *d-10* and *d-9*), and so on, moving across the limb until only *Hoxd-9* is expressed at the preaxial border (Dollé et al 1989). The fact that five different *Hox-d* expression domains can be distinguished prompted the suggestion that this may be why we have five fingers (and toes) (Tabin 1992): an attractive but unlikely explanation because, in the chick wing bud, the digits all come from the region that expresses all five *Hox-d* genes. The gene encoding the transcription factor *Lmx1* is expressed in the dorsal part of the limb in response to *Wnt 7a* signalling by dorsal ectoderm.

The molecular basis of limb development is conserved in humans. Mutations associated with many of the genes described above are responsible for human congenital limb malformations. Details can be found in Ferretti and Tickle (2006), Zeller et al (2009) and Zuniga et al (2012).

DEVELOPMENT OF LIMB TISSUES

DEVELOPMENT OF BONE AND CONNECTIVE TISSUE IN THE LIMB

The first signs of the formation of skeletal elements are regions of increased cell density in the core of the limb bud. The cells in these regions go on to differentiate into chondrocytes, which produce large quantities of extracellular matrix, in which they become embedded, giving rise to a cartilage model of the bone to be developed. One centre of chondrogenesis forms in the proximal region of an early human limb bud; two centres of chondrogenesis then form more distally as the limb bud lengthens, followed by five centres in the broadened distal tip. The *Sox9* transcription factor is essential for chondrocyte differentiation in mouse embryos; the *Sox9* gene is expressed before and during deposition of cartilage (Wright et al 1995). *Sox9* gene transcripts are expressed in the humerus and forearm skeletal elements of human embryos at approximately 44 days, and in the carpals, metacarpals and phalanges, in addition to the more proximal elements, at approximately 52 days (Fig. 15.4). Mutations in human *Sox9* are associated with campomelic dysplasia, a skeletal dysmorphology syndrome (Foster et al 1994). Some bony eminences, which are sites of entheses, have been shown to derive from chondrogenic foci which form separately and later than the cartilage model of the long bones (Blitz et al 2013). The cells co-express *Sox9* and the transcription factor *scleraxis* (*Scx*). Blockage of *Scx* or *Bmp4* expression in the limb arrests eminence development. It is suggested that as the fully formed eminences are subjected to tension whereas joint surfaces are subjected to compression, then different developmental pathways are in operation to cope with different functional parameters (Blitz 2013). *Scx* encoding the transcription factor

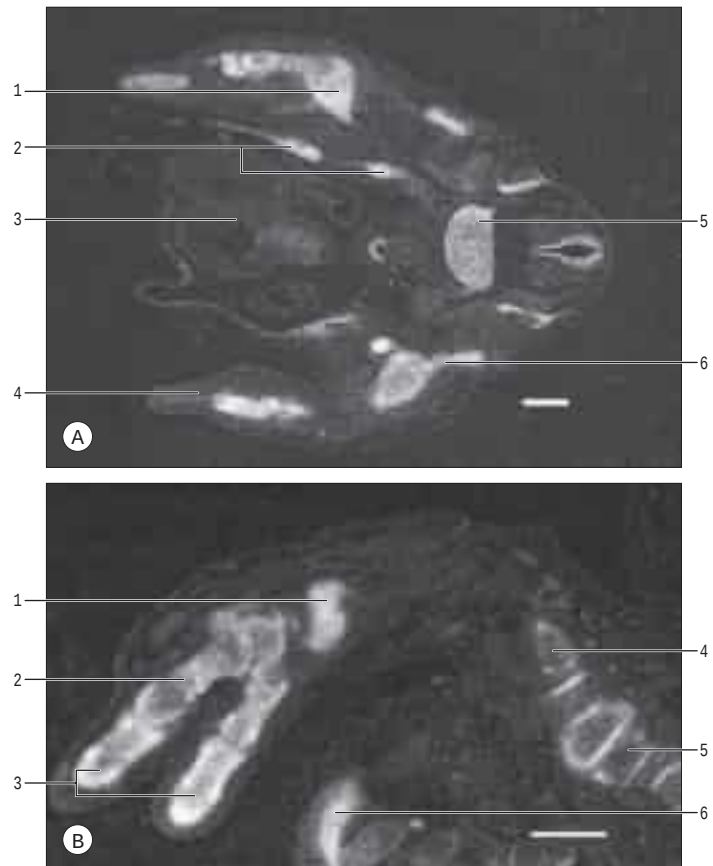


Fig. 15.4 *Sox9* expression in the skeleton of developing limbs of human embryos. **A**, Stage 18; approximately 44 days post ovulation. Key: 1, humerus; 2, ribs; 3, heart; 4, forearm; 5, vertebral body; 6, scapula. **B**, Stage 21; approximately 52 days post ovulation. Key: 1, carpal; 2, metacarpal; 3, phalanges; 4, humerus; 5, scapula; 6, rib. *Sox9* transcripts are detected in sections with radioactively labelled probes followed by autoradiography. Positive signal is represented by silver grains that show up as white areas. Scale bars represent 500 μ m in both A and B. (The human tissue was provided by the joint MRC-Wellcome Human Developmental Biology Resource (<http://www.hdbr.org>) at the Institute of Human Genetics, Newcastle upon Tyne, UK. By courtesy of Dr Susan Lindsay, Institute of Human Genetics, University of Newcastle upon Tyne, UK.)

DEVELOPMENT OF THE LIMBS

Scleraxis is expressed in the early limb subjacent to the ectoderm. Studies of *Scx* knockout mice show that the differentiation of force-transmitting tendons can be distinguished from muscle-anchoring tendons (Muchison et al 2007). A population of *Scx*⁺/*Sox9*⁺ progenitor cells have been shown to give rise to *Scx*⁺/*Sox9*⁺ chondrocytes and *Scx*⁺/*Sox9*⁺ tenocytes/ligamentocytes. The *Scx*⁺/*Sox9*⁺ cells contribute particularly to establishing the enthesis organ between the cartilage and tendon (Blitz et al 2013). The closer the tendon is to the cartilage primodium (anchoring tendon), the more tenocytes arise from the *Sox9*⁺ lineage. In general, the number of such derived tenocytes decreases with increasing distance from the cartilage model, although there is variation between individual force-transmitting tendons (Sugimoto et al 2013).

The first evidence of bone formation is seen at the mid section of the diaphyses of long bones at 7–8 weeks in human embryos. Vascular invasion of the cartilage matrix precedes the formation of a periosteal collar, and subsequently extends proximally and distally until it reaches the future epiphysal level, where a growth plate will be established. By 10 weeks, columns of chondrocytes can be seen at the epiphysal level of most bones. However, only the lower end of the femur and upper end of the tibia develop secondary ossification centres before birth.

For further details describing the development of cartilage, see page 84; for the development of bone, see page 91. For further details describing the development of specific limb bones, see the appropriate regional chapter. It should be noted that maternal nutrition and vitamin status affects fetal bone growth and that despite vitamin supplementation in Western countries, pregnant women in northern latitudes may remain vitamin D deficient in winter pregnancies. Infant length was shorter in pregnancies with early serum 25-hydroxyvitamin D below the median value (Walsh et al 2013).

DEVELOPMENT OF JOINTS IN THE LIMB

Regions of developing cartilage are easily recognized histologically in the developing limb because they consist of widely spaced cells surrounded by matrix. These regions are separated by transverse bands of relatively flattened cells – interzones – which mark the sites of future joints. Their subsequent development varies according to the type of joint that is formed.

In developing fibrous joints, the interzone is converted into collagen, which is the definitive medium connecting the bones involved. In developing synchondroses, the interzone becomes (growth) cartilage of the modified hyaline type; in developing symphyses, it is predominantly fibrocartilage. In developing synovial joints, the interzonal mesenchyme becomes trilaminar when a more tenuous intermediate zone appears, splitting the mesenchyme into two dense strata. As the skeletal elements chondrify, and in part ossify, the dense strata of the interzonal mesenchyme also become cartilaginous; subsequent cavitation of the intermediate zone establishes the cavity of the joint. The loose mesenchyme around the cavity forms the synovial membrane and probably also gives rise to all other intra-articular structures, such as ligaments, discs and menisci. In joints containing discs or menisci, and in compound articulations, more than one cavity may appear initially, sometimes merging later into a complex single cavity. As development proceeds, thickenings in the fibrous capsule can be recognized as the specializations peculiar to a particular joint. (In some joints, such as accessions to the fibrous capsule are derived from neighbouring tendons, muscles or cartilaginous elements.) Although the initial stages in the process of cavitation of joints are independent of movements, a true joint cavity can generally form only in the presence of movements.

DEVELOPMENT OF MUSCLE IN THE LIMB

It is now well established that all limb myogenic precursor cells originate from the somites (Chevallier et al 1977, Kardon et al 2003). These cells are committed at an early stage and can be identified in the lateral halves of the somites. After the mesenchymal sclerotome cells have migrated from the epithelial somite, the remaining dorsolateral portion is termed the epithelial plate or dermomyotome of the somite (see Fig. 44.3). Cells from the dorsomedial edge of the dermomyotomes form the axial musculature, whereas, at limb levels, cells de-epithelialize and migrate from the ventrolateral edges of the dermomyotome into the limb bud. These precursor limb myoblasts migrate through a non-random, structured network of extracellular fibrils. At their leading ends, the migrating cells exhibit filopodia, which are in contact with the extracellular fibrils or with other cells; it is believed that the orientation of the extracellular fibrils may direct the migration of the cells. The precursor muscle cells do not differentiate into muscle before their

migration into the limb, probably because of inhibitory signals produced by the somatopleuric mesenchyme. Muscle differentiation depends on the key activity of the MyoD family of transcription factors (Weintraub et al 1991).

The myogenic cells colonize the limb bud in a proximodistal direction but never reach the most distal portion of the limb. Experiments in chick embryos revealed that myogenic cells are still indifferent regarding their region-specific determination when they first enter the limb. Thus, when brachial-level somites are grafted opposite the leg-forming region, or when pelvic-level somites are grafted opposite the wing-forming region, a normal limb musculature develops, irrespective of the origin of the somites, i.e. the muscle cells, unlike the somatopleuric mesenchyme, are not specified to be wing or leg (Chevallier et al 1977). Other experiments show that the muscle pattern developed in the limb reflects the pattern of the skeletal elements: duplication or lack of digits is accompanied by the duplication or lack of the corresponding muscles (Robson et al 1994).

The first myogenic cells to arrive in the limb form the principal dorsal and ventral premuscle masses. It is believed that, in all classes of tetrapods, limb muscle development begins with these masses, which produce all the skeletal muscle in the limb. The premuscle masses undergo a stereotyped spatiotemporal sequence of divisions and subdivisions as the limb lengthens. In human embryos, this leads to the individualization of about 19 muscles in the upper limb and 14 muscles in the lower limb. The mechanisms that divide the muscle masses are not known; one suggestion is that these divisions are created by invading somatopleuric cells. In the upper limb, the premuscle masses first divide into three masses, the next division gives rise to the muscles attached to the carpus, and the final division produces the long muscles of the digits. A similar pattern is seen in the lower limb.

Each anatomical muscle appears as a composite structure. The muscle cells and myosatellite cells are of somitic origin, and the connective tissue envelopes and the tendons are of somatopleuric origin. The precise way in which the muscles become anchored to the developing bones by the tendons is not clear. For further development of skeletal muscle, see page 109.

The literature suggests that all musculoskeletal elements are in their appropriate positions in human limbs by 10 weeks. For detailed accounts of the process, consult O’Rahilly and Gardner (1975) and Uthoff (1990). For recent reviews on limb development, including tissue formation, see Lane and Tickle (2003).

DEVELOPMENT OF NERVES IN THE LIMB

During development, the spinal cord opposite the limbs becomes specialized, with the formation of longitudinal minicolumns (columels) along its rostrocaudal axis. The columels consist of groups of individual pools of differentiated motor neurones, precisely organized according to function (Jessell et al 2011). Motor neurones within each individual pool innervate a particular muscle. Neurones in medial pools express the transcription factor *Islet1* and their axons innervate ventral limb muscles, while neurones in lateral pools express the related transcription factor, *Lim1*, and innervate dorsal limb muscles. Neurones also express different Hox proteins according to their position along the rostrocaudal axis of the spinal cord, and this determines their identity and their position within the spinal cord (Dasen et al 2005).

Motor axons enter the developing limbs earlier than sensory axons at the time when dorsal and ventral muscle masses have started to form. Guidance of the growing axons is not dependent on the presence of migrating myoblasts; when these are absent, the main nerve trunks in the limb still form, although the smaller motor branches that would innervate individual muscles do not form (Lewis et al 1981, Honig et al 2005). Motor axons grow into the dorsal and ventral muscle masses and innervate muscle groups in a proximo–distal progression. Correct axonal routing depends on both repulsive and attractive cues. The pathfinding of motor axons is an important model system for understanding binary axon guidance decisions: glial cell (line)-derived neurotrophic factor (GDNF) attracts motor neurone growth cones, and interacts synergistically with ephrin-As to determine growth cone directionality (Drescher 2011). Experimental studies in mice and chickens have shown that axons of neurones in lateral pools grow into the dorsal part of the limb as a result of repulsive interactions between the receptor EphA4 (expressed by these motor neurones) and its ligand, ephrinA (expressed by ventral limb cells). The axons of EphB-expressing motor neurones in medial pools grow into the ventral part of the limb as a result of repulsive interactions with ephrinB-expressing cells in the dorsal limb (Drescher 2010). The dorso-ventral differences in Ephrin expression in the limb are controlled by the system that establishes the

dorso-ventral axis of the limb and that results in dorsal expression of *Lmx1*.

Sensory neurones in the dorsal root ganglia opposite the limbs send axons peripherally to terminate in the skin and to innervate proprioceptors in the limb muscles. The trajectories of sensory axons into the limb can be influenced by interaction with motor axons. When sensory axons grow into the developing limb, it has already been invaded by motor axons and the environment will have been altered by any extracellular factors they may have secreted. If motor axons are removed before sensory axons extend into the limb, those sensory axons, which would normally terminate in muscle (1a afferent axons), appear unable to do so and instead become cutaneous nerves (Honig et al 1986, Honig and Rutishauser 1996, Honig et al 1998).

The developing skin (ectoderm and immediately underlying somatopleuric mesenchyme) is essential for the normal development of cutaneous sensory axons. If the ectoderm is removed, the cutaneous nerves that would project to it are absent. This can be prevented by applying the signalling molecule BMP4 (Honig et al 2004, Honig et al 2005). Cutaneous sensory axons are not matched to particular regions of skin in the limb and can innervate regions outside their usual anatomical dermatomes, or expand into regions of skin where the cutaneous innervation is eliminated (Wang and Scott 2002).

Proprioceptive sensory neurones are characterized by expression of the transcription factor *Runx3* and also express a neurotrophic factor receptor (*TrkC*) that binds specifically to the neurotrophic factor NT-3. NT-3 is expressed in the limb bud and may serve as a short-distance guide to the ingrowing axons (Genc et al 2004). There is a reciprocal interaction with the neuronal soma: in the absence of NT-3, proprioceptive neurones in the dorsal root ganglia die. The central processes of the proprioceptive sensory neurones innervating limb muscle form monosynaptic connections to motor neurones in the spinal cord after their peripheral axons have extended into the limbs. However, the pattern for these sensory/motor connections appears to be specified early. The proximo-distal position of the sensory endings in the limb is linked to the dorso-ventral location of the target motor neurones in the spinal cord, such that proximal sensory axons will connect to ventral motor columns (Sürmeli et al 2011). NT-3 is also expressed in ventral neural tube and, as in the periphery, may help guide the axons to the correct location in the spinal cord.

The different types of sensory endings show activation at different developmental times; mechanoreceptors and proprioceptors are active ahead of nociceptive neurones prenatally. A third wave of mechanosensitivity acquisition by the remaining nociceptors occurs just after birth (Lechner et al 2009).

EMBRYONIC MOVEMENTS

Embryonic movements are vital for development of the musculoskeletal system (Pitsillides 2006, Nowlan et al 2010). Simple movements of an extremity have been observed sporadically as early as the seventh week of gestation in human embryos. Combined movements of limb, trunk and head commence between 12 and 16 weeks of gestation. Movements of the embryo and fetus encourage normal skin growth and flexibility, in addition to the progressive maturation of the musculoskeletal system.

FETAL MOVEMENTS

Fetal movements have been detected by ultrasonography in the second month of gestation. Those related to trunk and lower limb movements are perceived consistently by the mother from about 16 weeks' gestation (quickenings). Movements of the fetus often involve slow and asymmetric twisting and stretching movements of the trunk and limbs, which resemble athetoid movements. There may also be rapid, repetitive, wide-amplitude limb movements, similar to myoclonus.

By 32 weeks' gestation, symmetric flexor movements are most frequent. By term, the quality of the movements has generally matured to smooth, alternating movement of the limbs, with medium speed and intensity. The reduced effect of gravity *in utero* may cause certain fetal movements to appear, on ultrasonography, more fluent than the equivalent movements observed postnatally. The number of spontaneous

movements decreases after the thirty-fifth week of gestation and, from this time, there is an increase in the duration of fixed postures. This restriction of normal fetal movements in late gestation reflects the degree of compliance of the maternal uterus; there is a slowing of growth at this time.

In addition to promoting normal musculoskeletal development, movements of the fetus encourage skin growth and flexibility indirectly. Fetuses with *in utero* muscular dystrophies, or other conditions that result in small or atrophied muscles, have webs of skin, pterygia, which pass across the flexor aspect of the joints and severely limit movement. Multiple pterygium syndrome is characterized by webbing across the neck, the axillae and antecubital fossae. Usually, the legs are maintained straight and webbing is not seen at the hip and knee. A group of congenital disorders, collectively termed 'multiple congenital contractures', may result from genetic causes, or limitations of embryonic and fetal joint mobility, or may be secondary to muscular, connective tissue, skeletal or neurological abnormalities. These conditions may be recognized on prenatal ultrasound examination by the appearance of fixed, immobile limbs in bizarre positions, or by webbing in limb flexures. Specific syndromes, lethal multiple pterygium syndrome and congenital muscular dystrophy, have been described.

The workload undertaken by the musculoskeletal system before birth is relatively light because the fetus is supported by the amniotic fluid and, therefore, under essentially weightless conditions. The load on the muscles and bones is generated by the fetus itself, with little gravitational effect. The reduction of gravitational force afforded by the supporting fluid means that all parts of the fetus are subject to relatively equal forces and that the position assumed by the fetus relative to gravity is of little consequence. This is important to ensure the normal modelling of fetal bones, especially the skull. Skulls of premature babies may become distorted as a result of the weight of the head on the mattress, despite regular changes in position, and the application of oxygen therapy via a mask attached by a band around the head can cause dysostosis of the occipital bone.

VASCULATURE OF THE LIMB

ARTERIES

The early limb bud receives blood via intersegmental arteries that contribute to a primitive capillary plexus. At the tip of the limb bud, there is a terminal plexus, which is constantly renewed in a distal direction as the limb grows (see Figs 47.1, 79.1). Later, one main vessel, the axial artery, supplies the limb and the terminal plexus.

The development of the vasculature in the limb precedes the morphological and molecular changes that occur within the limb mesenchyme as tissues begin to form. Cartilage differentiation within the chick limb bud occurs only after local vascular regression begins, and only in areas with few or no capillaries (Hallmann et al 1987). It is not known whether the presence, or lack, of blood vessels affords different local environmental stimuli for mesenchymal cells (by varying the supply of nutrients to the tissue), or whether the local environment is controlled by the endothelial cells. Similarly, it is not clear whether inductive factors from the limb mesenchyme cause the changes that occur in the pattern of blood vessels. Work on chick wing buds suggests that the position of the central artery in the primitive limb bud vasculature depends on Shh signalling from the polarizing region (Davey et al 2007).

VEINS

At the tip of the early limb bud, blood in the terminal capillary plexus returns to the body via a marginal vein that develops along the pre- and postaxial borders of the limb. As the limb enlarges, the marginal vein can be subdivided into pre- and postaxial veins, which run along their respective borders and which are the precursors of the superficial veins of the limb. Generally, the preaxial (superficial) veins join the deep veins at the proximal joint, and the postaxial (superficial) veins join the deep veins at the distal joint of the limb. Deep veins develop *in situ* alongside the arteries.

KEY REFERENCES

- Ferretti P, Tickle C 2006 The limbs. In: Ferretti P, Copp A, Tickle C et al (eds) *Embryos, Genes and Birth Defects*, 2nd ed. Chichester: Wiley, pp. 123–66.
An account of the genetic basis of limb development.
- Hinchcliffe JR, Johnson DR 1980 *The Development of the Vertebrate Limb. An Approach through Experiment, Genetics, and Evolution*. Oxford: Clarendon Press.
A presentation of the classical experiments on limb development.
- Lane EB, Tickle C 2003 How to make a hand: a 3 day symposium held at the University of Dundee. *J Anat* 202:1.
A collection of reviews on limb development and human limb malformations.
- Nowlan NC, Sharpe J, Roddy KA et al 2010 Mechanobiology of embryonic skeletal development: insights from animal models. *Birth Defects Res C Embryo Today* 90:203–13.
An account that reviews and contrasts recent work on both chick and mouse models on the importance of movement in the development of the locomotor apparatus.
- O’Rahilly R, Gardner E 1975 The timing and sequence of events in the development of the limbs in the human embryo. *Anat Embryol* 148:1–23.
A description of the stages in human limb development.
- Pitsillides AA 2006 Early effects of embryonic movement: ‘a shot out of the dark’. *J Anat* 208:417–31.
A review of recent work using chicken embryos to explore the importance of movement in development of the locomotor apparatus.
- Uthoff HK 1990 *The Embryology of the Human Locomotor System*. Berlin: Springer.
A description of embryonic limb development.
- Zeller R, Lopez-Rios J, Zuniga A 2009 Vertebrate limb development; moving towards integrative analysis of organogenesis. *Nat Rev Genet* 10: 845–58.
A review of the molecular basis of limb development particularly focusing on how cell–cell interactions are integrated.
- Zuniga A, Zeller R, Probst S 2012 The molecular basis of human congenital limb malformations. *WIREs Dev Biol* 1:803–22.
A presentation of recently identified genes involved in limb development and their relevance to human congenital limb defects.

REFERENCES

- Blitz E, Sharir A, Akiyama H et al 2013 Tendon-bone attachment unit is formed modularly by a distinct pool of Scx- and Sox-positive progenitors. *Development* 140:2680–90.
- Chevallier A, Kieny M, Mauger A 1977 Limb-somite relationship: origin of the limb musculature. *J Embryol Exp Morphol* 41:245–58.
- Dasen JS, Tice BC, Brenner-Morton S et al 2005 A Hox regulatory network establishes motor neuron pool identity and target-muscle connectivity. *Cell* 123:477–91.
- Davey MG, James J, Paton IR et al 2007 Analysis of talpid3 and wild-type chicken embryos reveals roles for Hedgehog signalling in development of the limb bud vasculature. *Dev Biol* 301:155–65.
- Dollé P, Izpisua-Belmonte JC, Falkenstein H et al 1989 Coordinate expression of the murine Hox-5 complex homeobox-containing genes during limb pattern formation. *Nature* 342:767–72.
- Drescher U 2011 Axon guidance: push and pull with ephrins and GDNF. *Curr Biol* 21:R30–2.
- Dudley AT, Ros MA, Tabin CJ 2002 A re-examination of proximodistal patterning during vertebrate limb development. *Nature* 418:539–44.
- Fallon JF, Crosby GM 1977 Polarizing activity in limb buds of amniotes. In: Ede DA, Hinchliffe JR, Balls M (eds) *Vertebrate Limb and Somite Morphogenesis*. Cambridge: Cambridge University Press, pp. 55–70.
- Ferretti P, Tickle C 2006 The limbs. In: Ferretti P, Copp A, Tickle C et al (eds) *Embryos, Genes and Birth Defects*, 2nd ed. Chichester: Wiley, pp. 123–66.
An account of the genetic basis of limb development.
- Foster JW, Dominguez-Steglich MA, Guioli S et al 1994 Campomelic dysplasia and autosomal sex reversal caused by mutations in an SRY-related gene. *Nature* 372:525–30.
- Genc B, Ozdinler PH, Mendoza AE et al 2004 A chemoattractant role for NT-3 in proprioceptive axon guidance. *PLoS Biology* 12:e403.
- Hallmann R, Feinberg RN, Latker CH et al 1987 Regression of blood vessels precedes cartilage differentiation during chick limb development. *Differentiation* 34:98–105.
- Hinchliffe JR, Johnson DR 1980 *The Development of the Vertebrate Limb. An Approach through Experiment, Genetics, and Evolution*. Oxford: Clarendon Press.
A presentation of the classical experiments on limb development.
- Honig MG, Camilli SJ, Xue QS 2004 Ectoderm removal prevents cutaneous nerve formation and perturbs sensory axon growth in the chick hindlimb. *Dev Biol* 266:27–42.
- Honig MG, Camilli SJ, Surineni KM et al 2005 The contributions of BMP4, positive guidance cues, and repulsive molecules to cutaneous nerve formation in the chick. *Dev Biol* 282:257–73.
- Honig MG, Frase PA, Camilli SJ 1998 The spatial relationships among cutaneous, muscle neurons and motoneuron axons during development of the chick hindlimb. *Development* 125:995–1004.
- Honig MG, Lance-Jones C, Landmesser L 1986 The development of sensory projection patterns in embryonic chick hindlimb under experimental conditions. *Dev Biol* 118:532–48.
- Honig MG, Rutishauser UR 1996 Changes in the segmental pattern of sensory neuron projections in the chick hindlimb under conditions of altered cell adhesion molecule function. *Devel Biol* 175:325–37.
- Jessell TM, Sürmeli G, Kelly JS 2011 Motor neurons and the sense of place. *Neuron* 72:419–24.
- Kardon G, Campbell JK, Tabin C 2003 Local extrinsic signals determine muscle and endothelial cell fate and patterning in the vertebrate limb. *Dev Cell* 3:533–45.
- Lane EB, Tickle C 2003 How to make a hand: a 3 day symposium held at the University of Dundee. *J Anat* 202:1.
A collection of reviews on limb development and human limb malformations.
- Lechner SG, Frenzel H, Wang R et al 2009 Developmental waves of mechanosensitivity acquisition in sensory neuron subtypes during embryonic development. *EMBO J* 28:1479–91.
- Lewis J, Chevallier A, Kieny M et al 1981 Muscle nerve branches do not develop in chick wings devoid of muscle. *J Embryol Exp Morph* 64:211–32.
- MacCabe JA, Errick J, Saunders JW 1974 Ectodermal control of the dorsoventral axis of the leg bud of the chick embryo. *Dev Biol* 39:69–82.
- Muchison ND, Proce BA, Conner SDA et al 2007 Regulation of tendon differentiation by scleraxis distinguishes force-transmitting tendons from muscle-anchoring tendons. *Development* 134:2697–708.
- Niswander L, Tickle C, Vogel A et al 1993 FGF-4 replaces the apical ectodermal ridge and directs outgrowth and patterning of the limb. *Cell* 75:579–87.
- Nowlan NC, Sharpe J, Roddy KA et al 2010 Mechanobiology of embryonic skeletal development: insights from animal models. *Birth Defects Res C Embryo Today* 90:203–13.
An account that reviews and contrasts recent work on both chick and mouse models on the importance of movement in the development of the locomotor apparatus.
- O’Rahilly R, Gardner E 1975 The timing and sequence of events in the development of the limbs in the human embryo. *Anat Embryol* 148:1–23.
A description of the stages in human limb development.
- Parr BA, McMahon AP 1995 Dorsalizing signal Wnt-7a required for normal polarity of D-V and A-P axes of mouse limb. *Nature* 374:350–3.
- Pitsillides AA 2006 Early effects of embryonic movement: ‘a shot out of the dark’. *J Anat* 208:417–31.
A review of recent work using chicken embryos to explore the importance of movement in development of the locomotor apparatus.
- Riddle RD, Johnson RL, Laufer E et al 1993 Sonic hedgehog mediates the polarizing activity of the ZPA. *Cell* 75:1401–16.
- Robson LG, Kara T, Crawley A et al 1994 Tissue and cellular patterning of the musculature in chick wings. *Development* 120:1265–76.
- Saunders JW, Gasseling MT 1959 The differentiation of prospective thigh mesoderm grafted beneath the apical ectodermal ridge of the wing bud in the chick embryo. *Dev Biol* 1:281–301.
- Saunders JW, Gasseling MT 1968 Ectodermal-mesenchymal interactions in the origin of limb symmetry. In: Fleischmajer R, Billingham RE (eds) *Epithelial-mesenchymal Interactions*. Baltimore: Williams & Wilkins, pp. 78–97.
- Saunders JW, Gasseling MT, Errick JE 1976 Inductive activity and enduring cellular constitution of a supernumerary apical ectodermal ridge grafted to the limb bud of the chick embryo. *Dev Biol* 50:16–25.
- Sugimoto Y, Takimoto A, Akiyama H et al 2013 Scx⁺/Sox9⁺ progenitors contribute to the establishment of the junction between cartilage and tendon/ligament. *Development* 140:2280–8.
- Summerbell D, Lewis J, Wolpert L 1973 Positional information in chick limb morphogenesis. *Nature* 244:492–6.
- Sürmeli G, Akay T, Ippolito G et al 2011 Patterns of spinal sensory-motor connectivity prescribed by a dorsoventral positional template. *Cell* 147:653–65.
- Tabin CJ 1992 Why we have (only) five fingers per hand: hox genes and the evolution of paired limbs. *Development* 116:289–96.
- Tickle C, Barker H 2013 *The Sonic hedgehog gradient in the developing limb*. Wiley Interdiscip Rev Dev Biol 2:275–90.
- Uthoff HK 1990 *The Embryology of the Human Locomotor System*. Berlin: Springer.
A description of embryonic limb development.
- Walsh JM, Kilbane M, McGowan CA et al 2013 Pregnancy in dark winters: implications for fetal bone growth? *Fertil Steril* 99:206–211.
- Wang G, Scott SA 2002 Development of ‘normal’ dermatomes and somatotopic maps by ‘abnormal’ populations of cutaneous neurons. *Dev Biol* 251:424–33.
- Weintraub H, Davis R, Tapscott S et al 1991 The MyoD gene family: nodal point during specification of the muscle cell lineage. *Science* 251:761–6.
- Wright E, Hargrave MR, Christiansen J et al 1995 The Sry-related gene Sox9 is expressed during chondrogenesis in mouse embryos. *Nat Genet* 9:15–20.
- Zeller R, Lopez-Rios J, Zuniga A 2009 Vertebrate limb development; moving towards integrative analysis of organogenesis. *Nat Rev Genet* 10:845–58.
A review of the molecular basis of limb development particularly focusing on how cell-cell interactions are integrated.
- Zuniga A, Haramis AG, McMahon AP et al 1999 Signal relay by BMP antagonism controls the SHH/FGF-4 feedback loop in vertebrate limb buds. *Nature* 401:598–602.
- Zuniga A, Zeller R, Probst S 2012 The molecular basis of human congenital limb malformations. *WIREs Dev Biol* 1:803–22.
A presentation of recently identified genes involved in limb development and their relevance to human congenital limb defects.